PowerBI Tutorial

1. Import the supermarket sales dataset into Power BI and identify the data types of each column.
2. Create a **Card Visual** to display:
    - Total Sales
    - Total Quantity Sold
    - Average Customer Rating
3. Create a **Clustered Column Chart** showing **Total Sales by Branch**.
4. Create a **Bar Chart** showing **Total Sales by Product Line**.
5. Create a **Pie Chart** showing the percentage distribution of **Payment Methods**.
6. Create a **Donut Chart** showing the distribution of **Customer Type (Member vs Normal)**.
7. Create a **Column Chart** showing the **Total Quantity Sold by Product Line**.
8. Create a **Line Chart** to display **Daily Total Sales** using the Date field.
9. Create a **Line Chart** showing **Monthly Sales Trend**.
10. Create a **Stacked Column Chart** comparing **Sales by Branch and Customer Type**.
11. Create a **Stacked Bar Chart** showing **Product Line Sales by Gender**.
12. Create a **Treemap** displaying **Sales Contribution of Each Product Line**.
13. Create a **Scatter Chart** showing the relationship between **Unit Price and Quantity**, with bubble size representing Total Sales.
14. Create a **Histogram** (or grouped column chart using bins) for **Customer Ratings**.
15. Create a **Table Visual** displaying:

- Invoice ID
- Branch
- Product Line
- Quantity
- Total
- Payment

16. Create a **Matrix Visual** showing:

- Rows: Branch
- Columns: Product Line
- Values: Total Sales

17. Create a **Gauge Chart** comparing:

- Value: Average Rating
- Target: 8.0

18. Create a **Ribbon Chart** showing the ranking of **Product Lines by Sales across Branches**.

19. Create a **Funnel Chart** showing **Sales by Payment Method**.

20. Create a **Map Visual** displaying total sales by **City**.

21. Add a **Slicer** for:

- Branch
- City
- Product Line
- Payment Method
- Gender

22. Create a dashboard containing:

- Total Sales Card
- Sales by Branch
- Sales by Product Line
- Payment Method Pie Chart
- Daily Sales Trend
- Customer Rating Gauge
- Product Line Treemap

23. Format all charts by:

- Adding titles

- Displaying data labels
- Changing colours
- Adjusting legends
- Applying an appropriate theme

24. Create a report page that allows users to filter sales by **Branch** and **Product Line** using slicers.
25. Publish the completed Power BI report to the Power BI Service and verify that all visuals work correctly after publishing.